PROBLEM FS1

Melting Ice Cube

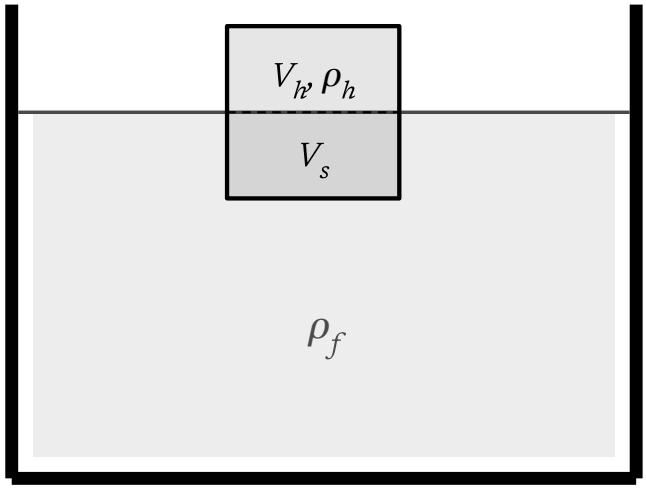
Original statement · fluid_statics_1.pdf

Archimedes | Floating body | Mass conservation

OFFICIAL PROBLEM STATEMENT

An ice cube of density ρ_h is partially submerged in a container holding a fluid of density $\rho_f > \rho_h$. The total volume of the ice cube is V_h, while the submerged part of the cube occupies a volume V_s to be determined. Calculate the volume of water displaced once the cube has melted, turning into water of density ρ_a, as a function of ρ_f, ρ_h, ρ_a and V_h, knowing that $\rho_h < \rho_a < \rho_f$. Particularize for the case $\rho_f = \rho_a$.

Translated from the Spanish original; the figure of the sheet labels the cube density ρ_b, the same quantity as ρ_h.



Statement figure (redrawn from fluid_statics_1.pdf, p. 1): ice cube of volume V_h and density ρ_h floating in a liquid of density $\rho_f > \rho_h$, submerged volume V_s

Source: fluid_statics_1.pdf (page 1, sheet of 11-2-00). Notes.pdf, Chapter 5: Archimedes' principle, Eq. 5.15.

Worked solution

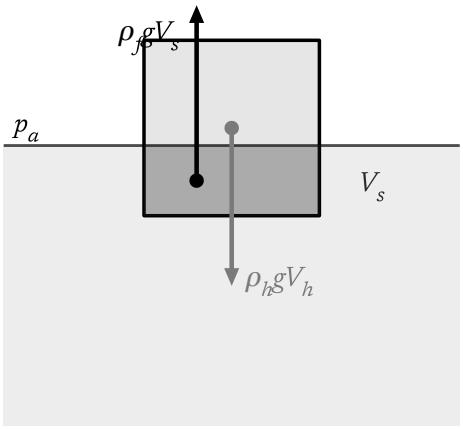
Hypotheses, Data and Unknowns

- The fluid and the melt water are at rest and incompressible; the air above them is at the uniform pressure p_a.
- The cube floats in equilibrium with part of its volume above the free surface (this requires $\rho_h < \rho_f$, given).
- Mass is conserved when the ice melts; the melt water and the fluid do not change volume when they meet (the melt water either floats as a layer, since $\rho_a < \rho_f$, or mixes with additive volumes).
- Data: ρ_h, ρ_f, ρ_a, V_h, g. Unknowns: the submerged volume V_s and the volume V_d that the liquid gains when the cube melts, that is the change of the volume of liquid below the free surface.

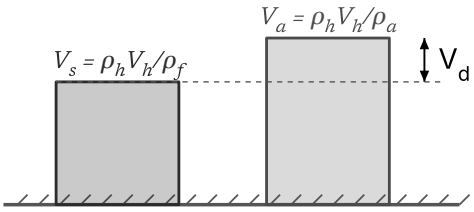
Symbol	Meaning	SI unit
ρ_h, ρ_f, ρ_a	densities of ice, fluid and melt water	kg/m^3
V_h	volume of the cube	m^3
V_s	submerged volume of the cube	m^3
V_a	volume of the melt water	m^3

The Floating Cube: Archimedes' Principle

The forces on the cube are its weight and the pressure of the fluid and of the air on its surface. The air pressure p_a acts over the whole closed surface of the cube and adds up to zero ($\oint p_a \vec{n}\, d\sigma = 0$). The part below the free surface feels the gauge pressure of the fluid, and by Archimedes' principle (Eq. 5.15) applied to the submerged part, closed by the plane of the free surface where the gauge pressure is zero, its resultant is the weight of the displaced fluid, $\rho_f g V_s$, upward.



(a) floating cube: weight and buoyancy



(b) after melting: $V_d = V_a - V_s$

(a) The floating cube: weight $\rho_h g V_h$ at its centre of gravity, buoyancy $\rho_f g V_s$ at the centroid of the submerged part. (b) The cube displaced V_s; once melted, the same mass occupies $V_a = \rho_h V_h / \rho_a$; the difference $V_d = V_a - V_s$ is the volume the liquid gains

Vertical equilibrium of the cube (upward positive):

$$\rho_f g V_s - \rho_h g V_h = 0 \quad \Longrightarrow \quad V_s = \frac{\rho_h}{\rho_f} V_h$$

Since $\rho_h < \rho_f$, $V_s < V_h$: part of the cube emerges, as assumed.

Melting: Conservation of Mass

When the ice melts, its mass $\rho_h V_h$ is conserved and becomes water of density ρ_a:

$$\rho_a V_a = \rho_h V_h \quad \Longrightarrow \quad V_a = \frac{\rho_h}{\rho_a} V_h$$

Before melting, the cube occupied the volume V_s below the free surface; after melting, the melt water occupies V_a in the liquid. The volume of liquid below the free surface therefore grows by

$$V_d = V_a - V_s = \rho_h V_h \left(\frac{1}{\rho_a} - \frac{1}{\rho_f} \right) = \frac{\rho_h}{\rho_a} V_h \left(1 - \frac{\rho_a}{\rho_f} \right)$$

In a container of horizontal section S, the free surface rises by V_d / S.

Answers and Checks

- Submerged volume: $V_s = (\rho_h / \rho_f)\, V_h$.
- Extra volume after melting: $V_d = (\rho_h / \rho_a)\, V_h\, (1 - \rho_a / \rho_f)$. With $\rho_a < \rho_f$ it is positive: the level rises, because the melt water is lighter than the fluid it displaced.
- Particular case $\rho_f = \rho_a$ (ice floating in its own water): $V_d = 0$. The melt water fills exactly the hole the cube had made, and the level does not change.
- Limiting case $\rho_h \to \rho_f$: $V_s \to V_h$, the cube is just fully submerged.
- Dimensions: V_d is a ratio of densities times V_h, so m^3; both brackets are dimensionless.
- Check against the official key: fluid_statics_1.pdf gives $V_s = (\rho_h/\rho_f)V_h$ and $V_d = (\rho_h/\rho_a)V_h(1 - \rho_a/\rho_f)$, the same results.